VFAST Transactions on Software Engineering
https://vfast.org/journals/index.php/VTSE@ 2023, ISSN(e): 2309-3978, ISSN(p): 2411-6246
Volume 11, Number 4, October-December 2023 *pp: 15-31*

Clusters of Success: Unpacking Academic Trends with K-Means Clustering in Education

Dua Agha [1], **Areej Fatemah Meghji** [1*], **Sania Bhatti** [1]

[1]Department of Software Engineering, Mehran University of Engineering and Technology Jamshoro, Pakistan

Keywords: *Clustering, Educational Data Mining, Machine Learning, Weka, K-Means, Student Performance*

Journal Info:
Submitted:
September 26, 2023
Accepted:
October 30, 2023
Published:
November 14, 2023

Abstract

Integrating Educational Data Mining (EDM) into the sector of education has heralded a new era, profoundly impacting learning outcomes by analyzing student performance and preventing academic disengagement. Using the K-Means clustering approach, this study carefully examines the academic accomplishments of students at Mehran University of Engineering and Technology and offers a sophisticated view of student performance patterns through the rigorous analysis of their learning data. A dataset comprising of the academic data of three student batches of the Department of Software Engineering was broken down into the subject categories of Computer Core, General, and Mathematics. The approach of clustering was then applied to find distinct performance patterns across the three subject categories. The findings of the research reveal that students have the highest performance in the computer core category, followed by mathematics, while the weakest overall performance across all three batches was exhibited in the general subject category. The study highlights the disparities in academic performance across distinct clusters and adds to our understanding of academic success while also illuminating the complex interactions between student characteristics and educational outcomes, providing useful information for educators and policymakers.

***Correspondence author email address:** areej.fatemah@faculty.muet.edu.pk
DOI: 10.21015/vtse.v11i4.1633

Introduction

As a fundamental human right, education is the cornerstone of personal development, social development, and national progress. The incorporation of data mining tools into the foundation of education has recently seen a revolutionary change. The way educational institutions operate has been revolutionized by the integration of Educational Data Mining (EDM), which uses vast data repositories to improve learning outcomes, forecast student performance, and prevent dropouts.

Education has historically evaluated student comprehension through interactions in the classroom, tests, and instructor feedback. However, the landscape of education has changed dramatically since the introduction of EDM. The multidisciplinary field of education and informatics known as EDM explores

This work is licensed under a Creative Commons Attribution 3.0 License.

extensive educational datasets using methods from data mining, machine learning, and statistics [1]. These techniques are employed to a diverse range of data sources, including student records, exam outcomes, participation in class, and online interactions [2]. Exploring these statistics, finding hidden trends, forecasting academic performance, and comprehending the learning behaviors of students are all possible for educators by utilizing the power of EDM.

Predictive modeling and learning analytics are two examples of cutting-edge research avenues that have emerged due to the implementation of EDM. Through the use of EDM and predictive modeling, educational institutions may forecast outcomes like dropout rates and identify internal factors that affect student performance [3]. To optimize learning environments and educational infrastructure, learning analytics, a subset of EDM, collects and analyzes student data [4]. Additionally, the use of EDM goes beyond evaluations of particular students. It is essential for forming educational policies, addressing challenges between teachers and students, and improving teaching strategies [5]. EDM enables customized teaching approaches, lowers dropout rates, and improves the overall quality of education by seeing patterns and trends in the data. In essence, EDM serves as a beacon, illuminating the path toward a more efficient, effective, and equitable education system.

In the vast landscape of machine learning, one fundamental technique that plays a pivotal role in uncovering intricate patterns within data is 'Clustering'. This method, essential in the realm of unsupervised learning, operates by grouping items based on their inherent similarities and identifies patterns in unlabeled data [6]. Unlike classification which works on labeled data, clustering utilizes algorithms to examine the information and find commonalities across data points, clustering similar occurrences together. The number of clusters may be set by the user or generated automatically. Similarities are measured using a variety of distance metrics. In addition to other things, this procedure helps with picture analysis, anomaly detection, and student segmentation. As clustering reveals hidden structures by examining raw data without previous labeling [7], it enables institutes to make data-driven decisions and get insightful knowledge from large datasets.

The main objective of this study is to meticulously examine the academic accomplishments of students of Mehran University of Engineering and Technology across different batches and academic disciplines. This initiative aims to assess students with a wide spectrum of academic capabilities and monitor their performance over the academic year. Through this, the study aims to extract valuable insights into the academic outcomes of the students. Employing clustering will facilitate the detection of concealed patterns and groups within the dataset. Consequently, this methodology strives to unveil subtle traits and trends that possess the potential to significantly impact students' academic achievements. This strategy aims to help individuals understand the complex interactions between academic achievement, student characteristics, and the educational environment by examining numerous aspects. The strategy utilized to forecast students' academic performance integrates the K-means clustering technique, a prevalent unsupervised machine learning method. K-means is instrumental in organizing data and identifying patterns, empowering institutions to derive valuable insights and make informed, data-driven decisions [8]. This process ultimately contributes to a more streamlined, proficient, and fair educational system.

The subsequent sections of the paper are structured as follows: the "Related Work" section offers an overview of pertinent literature. Following that, the "Research Methodology" section delineates the steps taken in this study, including data collection, preprocessing, educational data mining, clustering, and performance evaluation. The study's outcomes are then discussed in the "Results & Discussion" section. The limitations and potential future directions of the research are outlined in the section "Conclusion and Future Work".

Related Work

In recent years, the application of clustering algorithms in educational research has gained significant

attention, revolutionizing the way student performance is analyzed and understood. One prominent clustering algorithm utilized in various studies is the K-means algorithm, known for its ability to categorize data into distinct groups. In the realm of programming assessment, automatic programming assessment tools have been instrumental in evaluating students' programming abilities in C ++ Language. The data was collected from a score of 10 lab work assignments that had to be completed by each student. K-means clustering was used to classify 41 first-semester students into three groups based on their programming abilities: 16 with moderate abilities, 11 with high abilities, and 14 with lacking abilities. This approach simplifies the task of lecturers in managing and accessing student performance levels, enabling tailored learning experiences for students of different abilities [9]. Similar advancements have been observed in the domain of physical education, where researchers employed K-means clustering alongside support vector regression (SVR) to analyze college students' physical fitness levels at a university in China. The study utilized physical education test data from grade 14 students, considering variables such as weight, height, lung capacity, 50-meter run, long jump distance, and sitting position forward flexion [10].

Clustering techniques have also been pivotal in understanding students' academic performance and extracurricular activities. Research conducted at Sabaragamuwa University in Sri Lanka utilized data collected from 200 graduates via Google Forms consisting of 17 attributes covering personal information, educational information, other extracurricular activities, and hobbies. The approaches of K-means clustering, expectation-maximization, and hierarchical clustering were used to categorize students into clusters based on their academic and extracurricular activities. K-means outperformed other algorithms in categorizing students' performance levels [11].

In addition to academic performance, clustering methods have been applied to diverse datasets, such as those from the Oman Education Portal (OEP), to identify distinct performance profiles among students; the dataset consisted of 49,588 student records with 11 attributes of age, gender, religion, nationality, final marks, and so on, divided into four performance groups: "Excellent," "Good," "Average," and "Failure." Cluster analysis revealed distinct profiles for each group, with gender and age playing a significant role in performance, and concluded that female students have higher performance than male students [12].

K-means clustering has been instrumental in understanding the psychological aspects influencing students' motivation and interests. Research conducted at KSR PMI State University of Malang analyzed the impact of hidden curriculum and character building on self-motivation, employing K-means clustering to categorize students based on their motivation levels; the study found that hidden curriculum and character building significantly affect students' self-motivation, contributing 60.53% to self-motivation, with the remaining 39.47% influenced by other factors [13]. K-means clustering was also used to assess students' skills in specific domains, such as English language proficiency. A study analyzed the final scores of 30 English majors using the K-means algorithm, categorizing students into four clusters based on listening, speaking, reading, translation, and writing scores. It was found that cluster 1 (13% of students) excelled in all areas, cluster 2 (43%) had strong reading and translation skills but needed improvement in speaking, cluster 3 (33%) excelled in oral English but needed work on listening and writing, and cluster 4 (10%) had overall low scores, especially in listening. Utilizing K-means clustering, schools can identify individual student strengths and weaknesses, tailor teaching methods accordingly, and enhance teaching efficiency while optimizing resources [14].

The COVID-19 pandemic has further highlighted the importance of clustering techniques in understanding the educational landscape. Research analyzing the impact of the pandemic on higher education implemented K-means clustering to assess the disparities in online education access between urban and rural areas. The findings emphasized the digital divide, indicating the necessity for targeted initiatives to provide internet access to students in rural regions [15]. Additionally, K-means clustering

has been employed to identify suitable interest groups for higher education students comprising 176 data points that were analyzed. By categorizing students based on their interests revealed that 42.28% of students belonged to the Technical Club, 26.28% to the Communication Club, and 31.42% to the Sports Club [16].

Data mining with K-means clustering has played a crucial role in identifying students at risk of dropping out and facilitating timely interventions. Variables like semester credit system, quality, and Grade Predicate Academic (GPA) were used to analyze student data from the past four years [17]. In the context of academic performance prediction, K-means clustering has provided valuable insights for academic planners to monitor students' progression effectively. This study utilized a clustering model to analyze the academic performance of 79 students in a private institution in Nigeria. Using GPA and clustering with 3, 4, and 5 clusters, the study evaluated students' performance levels, enabling better decision-making and improvement strategies for subsequent academic sessions [18].

These research endeavors collectively underscore the transformative potential of K-means clustering in educational research, offering nuanced insights into student performance, motivation, and engagement. Using K-means this study aims to find patterns exhibited by high, average, and low performers within the subject categories of Computing Core, General, and Mathematics.

RESEARCH METHODOLOGY

This research aims to conduct a comprehensive analysis of students' academic performance across distinct batches and disciplines as shown in Figure 1. By considering students with varying academic traits and assessing their marks throughout the academic year, this study aims to gain valuable insights into the factors influencing their educational outcomes, particularly in computer core, general, and mathematics-based subjects. The goal is to identify underlying patterns and groupings within the dataset by using clustering analysis. Utilizing this methodology, the study seeks to identify characteristics and patterns that could help us better understand students' academic performance. The research intends to help educators comprehend the intricate relationships that exist between academic performance, student traits, and pedagogy.

1 Data Collection

This study utilized a dataset obtained from the examination department of Mehran University of Engineering and Technology (MUET), specifically concentrating on students of the Department of Software Engineering.

Table 1. Subject Category and Disciplines

Subject Category	Disciplines
Mathematics	Applied Calculus (AC)
	Applied Physics (AP)
	Linear Algebra and Analytical Geometry (LAAG)
Computer Core	Programming Fundamentals (PF)
	Object Oriented Programming (OOP)
	Introduction to Software Engineering (SWENG)
General	Introduction to Information & Communication Technologies (IICT)
	Functional English (FE)
	Pakistan Studies (PS)
	Professional Practices (PP)
	Islamic Studies / Ethics (IS-E)

The dataset encompasses information from 370 first-year students across three different batches. It incorporates data from three subject categories (computer-core, general, and mathematics), each of which includes several separate subjects spanning two semesters of education. For each subject, the dataset comprises of sessional marks, which consist of assignments and class tests, as well as midterm and final exam scores for every student. Subject categories and disciplines considered for the experiments are shown in Table 1.

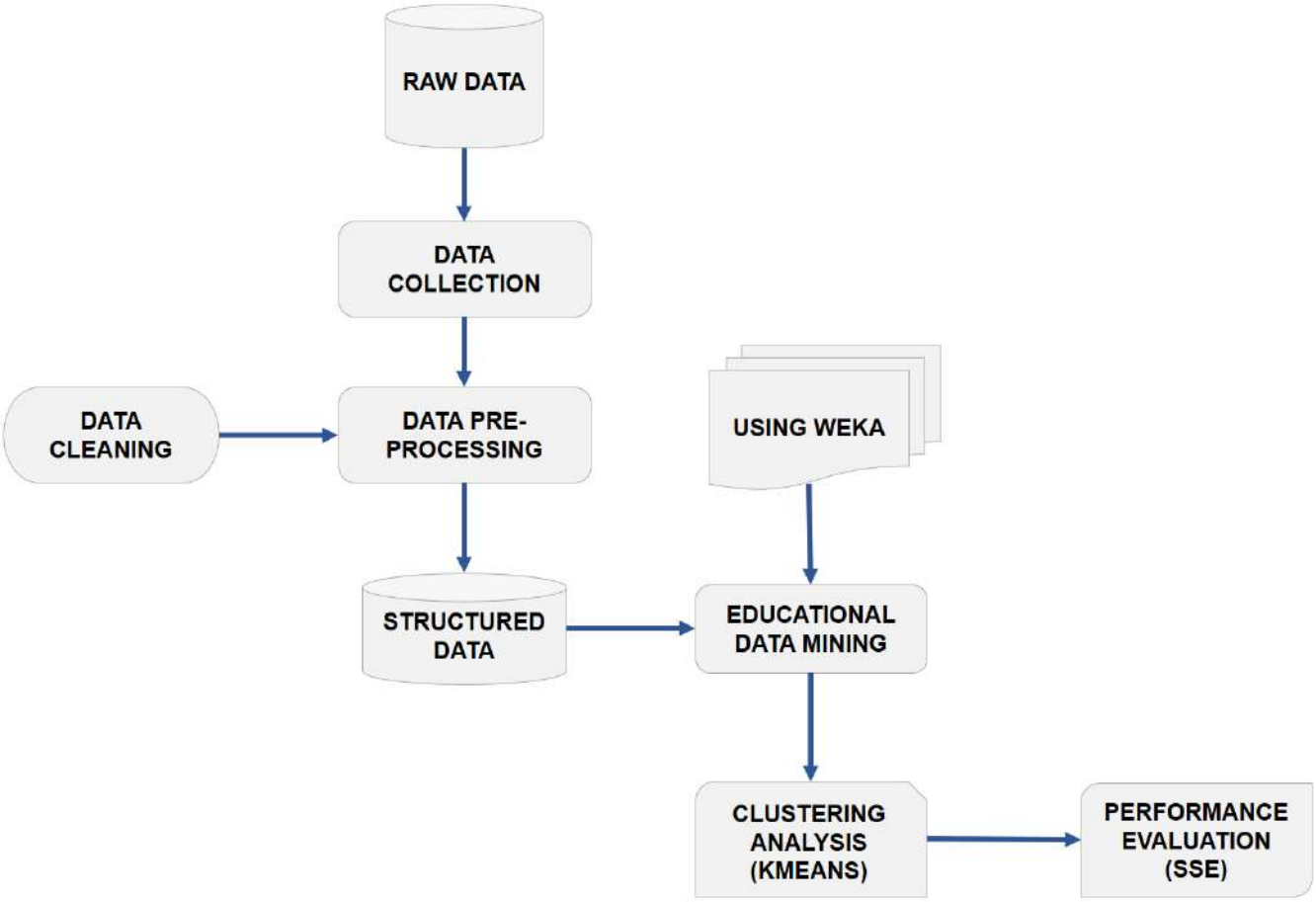


Figure 1. Research Methodology

2 Data Pre-Processing

Preparing and transforming raw data into an appropriate format for subsequent processing and analysis is the initial stage of data analysis. Often known as data preprocessing, this step ensures the dependability, homogeneity, and excellence of the data [1]. A series of steps were conducted to make the data more suitable for subsequent analytical tasks.

2.1 Data Cleaning

The dataset was carefully examined during the data preprocessing stage to ensure that no empty or missing values existed. The analysis can be significantly impacted by the existence of missing or null variables, thereby producing erroneous results [4]. Finding and filling in these missing values guarantees the dataset's integrity, which in turn guarantees the accuracy and dependability of subsequent results. The dataset was meticulously examined and improved after the data cleaning process concluded to assure its quality and relevance for additional study. This updated dataset, which now included 337 student records, was carefully refined to include accurate and complete data. To provide a more concentrated and reliable dataset suitable for in-depth research, a stringent filtering procedure was performed to remove any unnecessary or incomplete data points. The student batch and number of students considered for the experiment are shown in the Table. 2.

3 Educational Data Mining - Clustering

EDM is a field of research that combines data mining, machine learning, and statistical analysis to examine datasets. These datasets contain information re-

Table 2. Batch Student Distribution

Batch	No. of Students
Batch-1	101
Batch-2	102
Batch-3	133
Total	**337**

garding student performance, behavior, engagement metrics, instructional records, learning data, and teacher observations. The goal of EDM is to uncover hidden patterns within the data, offering valuable insights for educational stakeholders [4]. Without having prior knowledge of class labels, clustering is a fundamental unsupervised machine learning approach used to group data points that share similar traits or patterns. In a dataset, it seeks to find naturally occurring clusters based on data points that are more similar to one another than to those in other clusters [5].

The academic accomplishments of students from various batches and academic specialties were examined in-depth for this particular experiment. The study evaluated the academic growth of students with a range of academic qualities over an academic year. The main goal of this study is to identify underlying patterns in the student data to comprehend the student educational outcomes. K-means clustering was utilized to do this exploration.

3.1 K-MEANS

K-means is a popular unsupervised machine-learning method for data segmentation and pattern recognition. A dataset is divided into *'K'* unique, non-overlapping clusters; *K* is a user-defined parameter. The algorithm works by updating the centroids based on the mean of the data points in each cluster and iteratively allocating data points to the closest cluster centroid. K-means seeks to maximize separation between clusters while decreasing the sum of squared errors (SSE), which effectively reduces variation within each cluster [19].

The K-means algorithm involves several essential steps as highlighted in Figure 2. Initially, the algorithm determines the number of clusters, represented as *"k"*. Following this, centroids, which signify the center of each cluster, are initialized. The algorithm then computes the distance between data points and these centroids, typically utilizing a distance measure such as Euclidean distance, which calculates the straight-line distance between two points in space. It assess how close each data point is to the centroids, leading to the assignment of each point to the nearest cluster.

After this assignment, new centroids are calculated by averaging all points within each cluster. The algorithm iteratively repeats these steps while monitoring convergence and a maximum number of iterations. During each iteration, it evaluates the variance within clusters. Convergence in K-means signifies a state where centroids no longer significantly change between iterations. This indicates that the clustering has stabilized, and the algorithm has identified a suitable arrangement of data points into clusters. Achieving convergence is a crucial stopping criterion, preventing unnecessary iterations and identifying when the algorithm has reached a satisfactory solution.

The process continues until the lowest sum of variance is attained. Variance measures how spread-out data points are within each cluster. Lower variance implies that data points within a cluster are closely grouped around their centroid, indicating a more compact and well-defined cluster. Once the lowest variance is achieved, the clustering results are obtained, and data points are grouped into distinct clusters. This process provides valuable insights and patterns within the dataset.

4 Performance Evaluation

A crucial metric frequently used in statistical analysis and machine learning, particularly in the context of clustering algorithms, is SSE. The overall variance or dispersion of data points from their allocated cluster centroids is quantified by SSE. It is determined by adding together the squared Euclidean distances between each data point and the centroid of its corresponding cluster. SSE essentially assesses how compact a cluster is, with lower SSE values indicating clusters that are closer-knit and more cohesive [20].

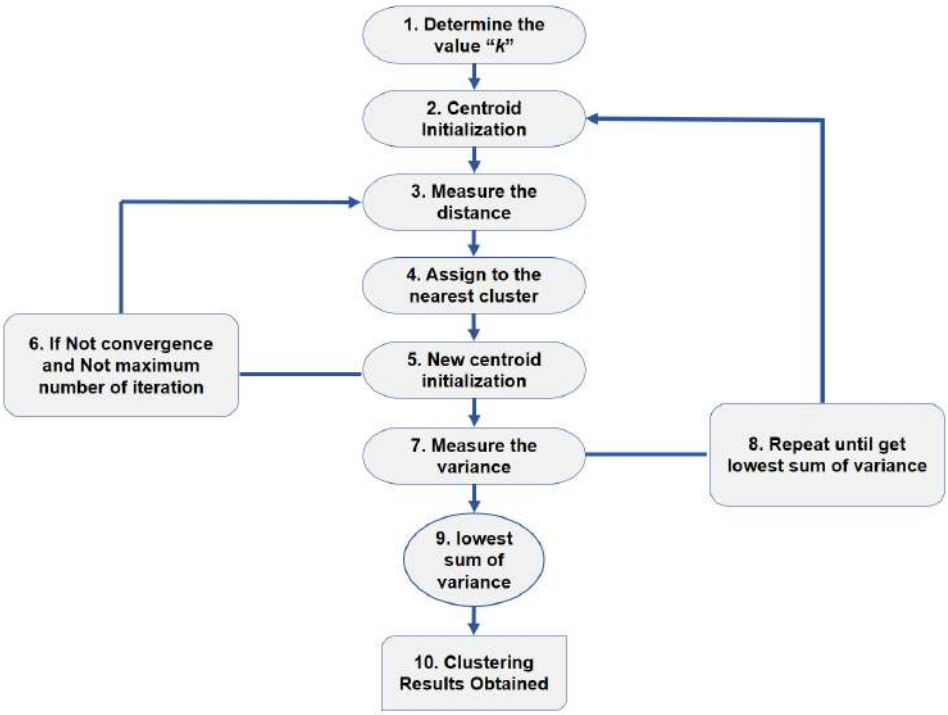


Figure 2. Working of K-Means Algorithm

SSE serves as a crucial tool for assessing the effectiveness and performance of clustering algorithms in research studies involving data segmentation and pattern recognition [21]. Equation 1 illustrates the mathematical representation of SSE:

$$SSE = \sum_{i=1}^{n} (x_i - \bar{x})^2 \quad (1)$$

Where n is the number of observations xi is the value of the ith observation.

5 RESULTS AND DISCUSSION

The results of the performed research employing K-means clustering on Batch 1, 2, and 3, encompassing the categories of Computing Core, General, and Mathematics have been provided in this section. Table 3 displays the K-means hyperparameters utilized in the experiments conducted using WEKA.

In the research study involving Computer Core subjects for Batch-1 (Table 4), K-means clustering analysis was conducted for a group of 101 students based on their performance in various assessments. Three distinct clusters emerged from the analysis

- CLUSTER 1: These students demonstrated moderate performance across all assessments.
- CLUSTER 2: This group exhibited the highest performance across assessments, particularly in final examinations.
- CLUSTER 3: Students in this cluster scored lower on average, especially in final examinations.

Focusing on computer core subjects for Batch-2, K-means clustering analysis was performed on 102 students based on their academic performance across various assessments (Table 5). The following three clusters emerged:

- CLUSTER 1: Students in this cluster displayed consistent and balanced performance across subjects like PF, SWENG, and OOP, maintaining moderate scores in sessionals, mid-term exams, and final exams.
- CLUSTER 2: This group consisted of high-achieving students who excelled in final examinations across all subjects, while also performing well in sessionals and mid-term exams.
- CLUSTER 3: Students in this cluster demonstrated overall lower academic performance,

Table 3. K-means Hyperparameters

Hyperparameters	Value Assigned
canopyMaxNumCanopiesTo HoldinMemory	100
canopyMinimumCanopyDensity	2
canopyPeriodic PruningRate	10000
canopyT1	-1.25
canopy T2	-1
debug	FALSE
distanceFunction	EuclideanDistance -R first - last
initializationMethod	Random
maxIterations	500
numClusters	3
numExecutionSlots	1
preserveInstancesOrder	FALSE
seed	10

particularly struggling in final exams.

Focusing on Batch-3, K-means clustering analysis was conducted to categorize 133 students based on their academic performance in various assessments. The following three clusters were identified in the experiment:

- CLUSTER 1: This group displayed consistently strong academic performance across subjects like PF, SWENG, and OOP, achieving high scores in sessionals, mid-term exams, and final exams.
- CLUSTER 2: Comprising 51% of the students, this cluster consisted of high-achievers who excelled in sessionals and mid-term exams but showed slightly lower scores in final examinations, particularly in OOP.
- CLUSTER 3: Representing a smaller proportion (9%) of students, this cluster exhibited significantly lower academic performance across all assessments, especially in final exams.

5.1 Experiment#2 - General

Focusing on general subjects (see Table 7), the three clusters generated using K-means for Batch-1 had the following main features.

- CLUSTER 1: Students in this cluster consistently achieved higher scores across assessments in various subjects.
- CLUSTER 2: This group displayed mixed performance, with stronger results in certain assessments and subjects.
- CLUSTER 3: Students in this cluster generally scored lower in assessments across all subjects.

Focusing on general subjects for Batch-2 having 102 students the following clusters were identified (see Table 8):

- CLUSTER 1: These students displayed consistent and moderate performance across all subjects and assessments, maintaining balanced scores.
- CLUSTER 2: The largest cluster (65% of students) consisted of high-achievers who excelled in final examinations across all subjects while performing well in sessionals and mid-term exams.
- CLUSTER 3: Representing 21% of students, this cluster exhibited slightly lower performance across assessments, particularly in final exams.

Performing K-means clustering analysis on Batch-3 comprising 133 students (see Table 9), the following clusters were identified:

- CLUSTER 1: This group consisted of students who consistently excelled in sessionals, mid-term exams, and final exams across multiple subjects, showcasing a strong academic aptitude.
- CLUSTER 2: Students in this cluster achieved moderately high academic performance but with slightly lower scores compared to CLUSTER 1. They performed well in sessionals and mid-term exams but showed a slight dip in final exam scores.
- CLUSTER 3: Representing 11% of the students, this cluster displayed significantly lower academic performance across all assessments when compared to the other clusters.

Table 4. Batch 1 – Computer Core Subjects

Batch 1 – Computer Core Subjects	Full Data	Cluster 1 High Performance	Cluster 2 Average Performance	Cluster 3 Low Performance
PF-Sessionals-s1	14.1287	15	14.0811	13.2
PF-Mid Exam Marks-s1	15.6832	17.0588	16.4595	13.1667
PF-Final Exam Marks-s1	44.4059	48.1176	46.3784	37.7667
SWENG-Sessionals-s2	18.495	18.7647	18.6757	17.9667
SWENG-Mid Exam Marks-s2	17.7921	18.0294	17.9459	17.3333
SWENG-Final Exam Marks-s2	38.9604	41.9118	43.0811	30.5333
OOP-Sessionals-s2	16.7327	16	18	16
OOP-Mid Exam Marks-s2	16.802	16.6471	18	15.5
OOP-Final Exam Marks-s2	44.0198	46.5588	49	35
Cluster Percentage (%)		34	37	30
Total No. of Students	101			
Number of Iterations	3			
Sum of Squared Errors	23.626			

Table 5. Batch 2 – Computer Core Subjects

Batch 2 – Computer Core Subjects	Full Data	Cluster 1 High Performance	Cluster 2 Average Performance	Cluster 3 Low Performance
PF-Sessionals-s1	15.0196	15.0323	15.8958	13.1739
PF-Mid Exam Marks-s1	15.8725	15	17.6042	13.4348
PF-Final Exam Marks-s1	48.0686	47.3548	51.875	41.087
SWENG-Sessionals-s2	12	11.8065	14.625	6.7826
SWENG-Mid Exam Marks-s2	15.2843	15.2581	15.6875	14.4783
SWENG-Final Exam Marks-s2	45.951	46.1613	48.3958	40.5652
OOP-Sessionals-s2	17.4608	17.2258	17.9375	16.7826
OOP-Mid Exam Marks-s2	13.9608	13.9677	16.2708	9.1304
OOP-Final Exam Marks-s2	41.0784	38.0968	48.9375	28.6957
Cluster Percentage (%)		31	48	23
Total No. of Students	102			
Number of Iterations	7			
Sum of Squared Errors	22.785			

Table 6. Batch 3 - Computer Core Subjects

Batch 3 - Computer Core Subjects	Full Data	Cluster 1 High Performance	Cluster 2 Average Performance	Cluster 3 Low Performance
PF-Sessionals-s1	15.4135	16.274	14.451	13.8889
PF-Mid Examination Marks-s1	16.4135	17.5342	15.2157	14.1111
PF-Final Examination Marks-s1	44.3083	47.3014	40.8431	39.6667
SWENG-Sessionals-s2	16.2331	16.4658	16.2157	14.4444
SWENG-Mid Examination Marks-s2	15.3383	16.0685	15.0196	11.2222
SWENG-Final Examination Marks-s2	48.2632	50.9863	46.9412	33.6667
OOP-Sessionals-s2	16.5789	18.863	15.549	3.8889
OOP-Mid Examination Marks-s2	12.4286	15.2877	9.8431	3.8889
OOP-Final Examination Marks-s2	40.3985	47.6164	35.6471	8.7778
Cluster Percentage (%)		73	51	5
Total No. of Students	133			
Number of Iterations	12			
Sum of Squared Errors	20.0882			

Table 7. Batch 1 - General Subjects

Batch 1 - General Subjects	Full Data	Cluster 1 High Performance	Cluster 2 Average Performance	Cluster 3 Low Performance
PP-Sessionals-s2	16.5743	17.1818	16.25	15.92
PP-Mid Exam Marks-s2	14.3069	15.6818	13.4063	13.04
PP-Final Exam Marks-s2	34.7822	42.6364	32.8438	23.44
FE-Sessionals-s1	17.7525	18.2273	17.375	17.40
FE-Mid Exam Marks-s1	16.6733	17.9773	16.375	14.76
FE-Final Exam Marks-s1	46.703	51.2955	45.9688	39.56
IS-E-Sessionals-s2	9.6931	9.9545	9.9688	8.88
IS-E-Mid Exam Marks-s2	7.5545	8.0455	7.3438	6.96
IS-E-Final Exam Marks-s2	17.1485	20.7955	16.2813	11.84
PS-Sessionals-s2	8.4158	8.2955	8.625	8.36
PS-Mid Exam Marks-s2	8.0198	8.2955	7.875	7.72
PS-Final Exam Marks-s2	22.9307	24.1364	22.5938	21.24
IICT-Sessionals-s1	9.0594	9.3409	9.3125	8.24
IICT-Mid Exam Marks-s1	7.2079	7.9318	5.375	8.28
IICT-Final Exam Marks-s1	21.1188	23.6818	17.9688	20.64
Cluster Percentage (%)		44	32	25.00
Total No. of Students	101			
Number of Iterations	5			
Sum of Squared Errors	46.403			

Table 8. Batch 2 - General Subjects

Batch 2 - General Subjects	Full Data	Cluster 1 Average Performance	Cluster 2 High Performance	Cluster 3 Low Performance
PP-Sessionals-s2	16.7157	16.75	16.9385	16
PP-Mid Examination Marks-s2	14.8824	14	16.1077	11.7619
PP-Final Examination Marks-s2	41.0196	34.875	45.1077	33.0476
FE-Sessionals-s1	16.1275	16.0625	16.3077	15.619
FE-Mid Examination Marks-s1	14.7941	13.5	15.4154	13.8571
FE-Final Examination Marks-s1	43.451	41.875	45.2308	39.1429
IS-E-Sessionals-s2	9.4902	9.5625	9.8615	8.2857
IS-E-Mid Examination Marks-s2	6.9804	6	7.5231	6.0476
IS-E-Final Examination Marks-s2	20.7843	18	22.6769	17.0476
PS-Sessionals-s2	9.0392	8.6875	9.0308	9.3333
PS-Mid Examination Marks-s2	7.7647	7.3125	8.1846	6.8095
PS-Final Examination Marks-s2	24.1863	23.8125	24.7231	22.8095
IICT-Sessionals-s1	8.2941	5.875	8.7846	8.619
IICT-Mid Examination Marks-s1	7.6176	6.625	8.1692	6.6667
IICT-Final Examination Marks-s1	22.2255	21.1875	23.4	19.381
Cluster Percentage (%)		16	65	21
Total No. of Students	102			
Number of Iterations	10			
Sum of Squared Errors	50.701			

Table 9. Batch 3 - General Subjects

Batch 3 - General Subjects	Full Data	Cluster 1 High Performance	Cluster 2 Average Performance	Cluster 3 Low Performance
PP-Sessionals-s2	16.7519	17.593	15.9444	12.8182
PP-Mid Examination Marks-s2	15.0602	16.8488	12.9167	8.0909
PP-Final Examination Marks-s2	42.4436	48.7558	37.8889	8
FE-Sessionals-s1	20	20	20	20
FE-Mid Examination Marks-s1	12.8195	13.3721	11.5278	12.7273
FE-Final Examination Marks-s1	38.9624	41.1395	34.4167	36.8182
IS-E-Sessionals-s2	9.8346	10	9.9722	8.0909
IS-E-Mid Examination Marks-s2	6.8947	7.3721	6.2222	5.3636
IS-E-Final Examination Marks-s2	21.8195	23.6163	19.2778	16.0909
PS-Sessionals-s2	8.0827	8.1047	8.0556	8
PS-Mid Examination Marks-s2	7.6917	8.0116	7.4167	6.0909
PS-Final Examination Marks-s2	22.6316	23.5465	22.1389	17.0909
IICT-Sessionals-s1	6.3609	7.093	4.5278	6.6364
IICT-Mid Examination Marks-s1	7.7895	8.3372	6.4444	7.9091
IICT-Final Examination Marks-s1	21.8045	22.9884	19.4444	20.2727
Cluster Percentage (%)		86	36	11
Total No. of Students	133			
Number of iterations	3			
Sum of Squared Errors	36.772			

5.2 Experiment#3 - Mathematics

As observed in Table. 10, focusing on the performance of students in mathematics-related subjects K-means clustering analysis for Batch-1 uncovered the following three clusters

- CLUSTER 1: This group demonstrated consistently strong academic performance across subjects and assessments, excelling in sessionals, mid-term exams, and final exams.
- CLUSTER 2: Students in this cluster exhibited higher performance levels in sessionals and mid-term exams, but slightly lower scores in final exams, particularly in subjects like AP and AC.
- CLUSTER 3: This cluster represented students with overall lower academic performance, struggling across all assessments and notably performing poorly in LAAG final exams.

For Batch-2 (see Table 11), K-means clustering analysis identified the following three clusters:

- CLUSTER 1: This group exhibited consistently strong academic performance across subjects like AP, AC, and LAAG, achieving high scores in sessionals, mid-term exams, and final exams.
- CLUSTER 2: Comprising 34% of the students, this cluster consisted of high-achievers who excelled in sessionals and mid-term exams but showed slightly lower scores in final examinations, especially in AP and AC.
- CLUSTER 3: Representing a small portion (4%) of students, this cluster displayed an overall lower academic performance, with notable weaknesses in final exams across all subjects.

Similarly focusing on mathematics-related subjects for Batch-3 (see Table 12), the following distinct clusters were identified:

- CLUSTER 1: This group consisted of students who maintained consistent and balanced academic performance across subjects like AP, AC, and LAAG, achieving moderate scores in sessionals, mid-term exams, and final exams.
- CLUSTER 2: Comprising 27% of the students, this cluster consisted of high-achievers who excelled in final examinations across all subjects, while also performing well in sessionals and mid-term exams.
- CLUSTER 3: Representing the largest portion (54%) of students, this cluster exhibited varying academic performance, with notably lower scores in final exams, particularly in AP and LAAG

6 Discussion

Heatmaps are frequently used in data visualization to quickly and intuitively reveal insights into complex datasets, enabling users to swiftly observe patterns and trends in the data being displayed [22]. The shades of red, white, and blue are used to represent outcomes in a range of intensities, from positive to negative. In particular, these hues act as visual clues to aid in the interpretation of data, with red often signifying positive or good outcomes, white signifying moderate or intermediate outcomes, and blue signifying negative or subpar outcomes.

Figure 3 represents the heatmap for the clusters of the current study, where Figure 3a) represents Computer Core, Figure 3b) represents General, and Figure 3c) represents the student clusters for Mathematics. We can observe from a closer look at the heatmap that the students have the best overall performance in the Computer Core category. We can see from Figure 3a), student performance is exceptional in the subjects of PF and SWENG. However, student performance is not satisfactory in OOP; B3-cluster 3 has the worst performance in this subject in the sessional, mid, and final examinations. Students seem to be doing well in the Mathematics category as well as can be observed from Figure 3c). The best student performance is in the subject AP followed by AC. The performance in the subject LAAG needs improvement; students across all clusters seem to be struggling to score high in this subject. The worst performance by the students is in the General category, with students only having satisfactory performance in the subject FE. Student performance for all batches is beyond satisfactory in the subjects IS, PS, and IICT.

The findings of this study highlight important trends that were discovered by utilizing the K-means

Table 10. Batch 1 – Mathematics

Batch 1 – Mathematics	Full Data	Cluster 1 High Performance	Cluster 2 Average Performance	Cluster 3 Low Performance
AP-Sessionals-s1	16.5347	18.9333	15.1026	16.0313
AP-Mid Examination Marks-s1	16.5248	18.6333	15.7949	15.4375
AP-Final Examination Marks-s1	37.1485	45.1	37.7436	28.9688
AC-Sessionals-s1	18.6634	18.9	18.6923	18.4063
AC-Mid Examination Marks-s1	16.505	18.1	16.2051	15.375
AC-Final Examination Marks-s1	35.1584	40.5667	34.9487	30.3438
LAAG-Sessionals-s2	18.2574	18.9	18.7949	17
LAAG-Mid Examination Marks-s2	16.3267	17.1667	16.6667	15.125
LAAG-Final Examination Marks-s2	36.8317	49.4	43.9744	16.3438
Cluster Percentage (%)		30	39	32
Total No. of Students	101			
Number of iterations	9			
Sum of Squared Errors	31.071			

Table 11. Batch 2 – Mathematics

Batch 2 – Mathematics	Full Data	Cluster 1 High Performance	Cluster 2 Average Performance	Cluster 3 Low Performance
AP-Sessionals-s1	17.3824	18.3906	16.0588	12.5
AP-Mid Examination Marks-s1	13.9118	15.7031	11.7059	4
AP-Final Examination Marks-s1	42.5686	46.7344	36.7647	25.25
AC-Sessionals-s1	17.9608	18.2188	17.2941	19.5
AC-Mid Examination Marks-s1	16.4216	17.875	14.0588	13.25
AC-Final Examination Marks-s1	38.4902	40.5938	34.0882	42.25
LAAG-Sessionals-s2	18.7451	18.7656	18.7059	18.75
LAAG-Mid Examination Marks-s2	17.0196	17.8125	16.0294	12.75
LAAG-Final Examination Marks-s2	43.2647	47.875	37.1471	21.5
Cluster Percentage (%)		64	34	4
Total No. of Students	102			
Number of iterations	5			
Sum of Squared Errors	50.0144			

Table 12. Batch 3 – Mathematics

Batch 3 – Mathematics	Full Data	Cluster 1 High Performance	Cluster 2 Average Performance	Cluster 3 Low Performance
AP-Sessionals-s1	17.0376	17.1346	16.6296	17.1481
AP-Mid Examination Marks-s1	13.8421	16.1538	10.2963	13.3889
AP-Final Examination Marks-s1	40.5564	45.1731	34.2593	39.2593
AC-Sessionals-s1	18.7519	19.4038	18.4074	18.2963
AC-Mid Examination Marks-s1	13.203	15.9423	14.7037	9.8148
AC-Final Examination Marks-s1	40.391	50.3462	43.5185	29.2407
LAAG-Sessionals-s2	17.188	18.1923	14.4444	17.5926
LAAG-Mid Examination Marks-s2	14.594	16.8654	9.8519	14.7778
LAAG-Final Examination Marks-s2	42.2782	50	25.5556	43.2037
Cluster Percentage (%)		52	27	54
Total No. of Students	133			
Number of iterations	9			
Sum of Squared Errors	31.35			

clustering technique with heatmap analysis. Batch 3 has performed well in the Computer Core category as well as the Mathematics category across a majority of the subjects. Batch 2 exhibited a lower performance, placing third in the evaluation. In our comprehensive experiment spanning General, Computer Core, and Mathematics categories consisting of multiple subjects, we meticulously analyzed varied clusters with Cluster-1 showcasing outstanding performance, showing that students in this cluster have a deep understanding across all categories. Cluster-2 showed average performance, demonstrating competence but not extraordinary mastery and Cluster-3 showed lower performance levels, revealing possible areas for improvement.

Additionally, our analysis revealed that the increase in student population coincided with a decline in the Sum of Squared Errors (SSE), indicating an increased correlation of the data to the clusters. This statistical measure underscored the reliability of our findings, reinforcing the correlation between a growing student populace, enhanced academic performance, and the distinct performance patterns within the identified clusters.

There are certain limitation to the current study. Firstly, our analysis was confined to data from first-year students, which means our findings might differ if we had examined longitudinal data. First-year students typically start off unfamiliar with the subjects, gradually adapting to the academic system over time, which could lead to improved performance as they progress. Secondly, our research revealed that students performed well in final exams but showed a need for greater effort in mid-term exams and sessional assessments. However, the specific factors contributing to the challenges in mid-terms and sessionals can further be explored. These limitations highlight areas for further research and analysis to gain a comprehensive understanding of student performance dynamics.

As a whole, our research not only highlights disparities in academic performance across distinct clusters but also uncovers the crucial influence of a larger student population on fostering academic excellence. These findings offer valuable insights for educators and policymakers, highlighting the need for tailored interventions to help underachieving students while simultaneously nurturing and challenging high-achievers to realize their full potential.

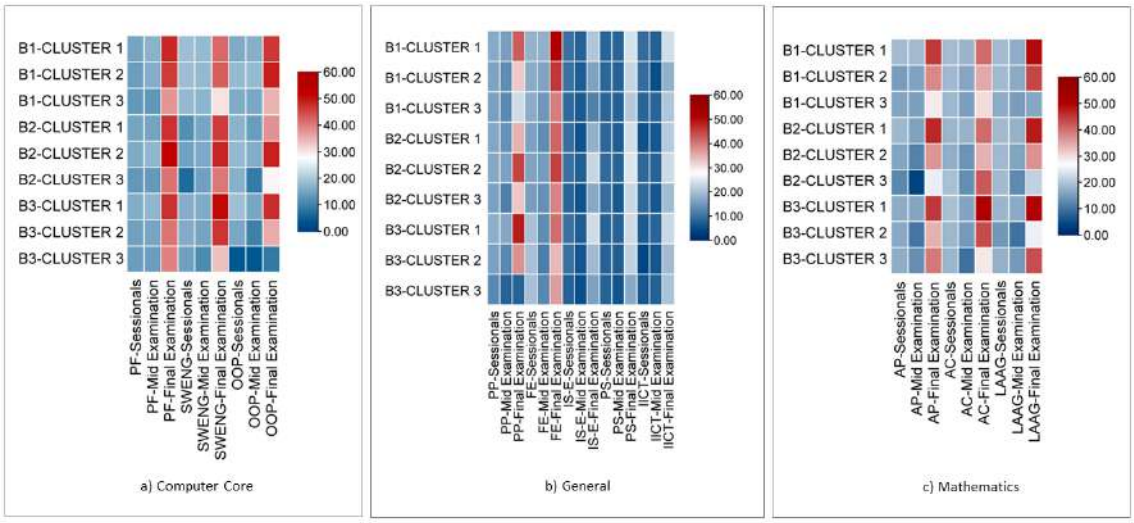


Figure 3. Heat Map Analysis: Computer Core Category (3a) – General (3b) – Mathematics (3c)

7 CONCLUSION AND FUTURE WORK

This study delves into the academic performance of students, analyzing their marks over the academic year to unravel crucial insights into the factors shaping their educational outcomes. Employing the clustering technique of K-means clustering, evaluated based on the metric of SSE, this research aims to uncover intricate patterns within the dataset. Performing heatmap analysis on the generated clusters revealed that students have the best overall performance in the Computer Core category, with the worst performance in the General subject category. In particular, Batch-1 and Batch-3 have better performance in the computer core category as compared to Batch-2. It was observed that students in every batch have better performance in final exams but face challenges in sessional and mid-term assessments, resulting in a decline in their overall academic performance. These findings not only increase our comprehension of the intricate relationships among academic performance, student characteristics, and educational contexts but also provide valuable insights for educators and policymakers striving to improve student's learning experiences.

Expanding datasets to incorporate qualitative data, such as student attitudes and learning styles, should be considered in future academic success research to provide a thorough knowledge of the factors influencing academic accomplishments. In addition, analyzing the effects of extracurricular activities, parental involvement, and socioeconomic variables in conjunction with identified clusters might help inclusive methods of learning. These efforts will refine our understanding of academic success dynamics and contribute to targeted educational strategies for diverse student demographics.

Author Contributions

Dua Agha: Methodology, Software, Visualization, Original draft preparation. **Areej Fatemah Meghji:** Conceptualization, Methodology, Supervision, Validation, Reviewing, and Editing. **Sania Bhatti:** Supervision, Validation.

Compliance with Ethical Standards

It is declared that all authors don't have any conflict of interest. Furthermore, informed consent was obtained from all individual participants included in the study.

References

[1] A. S. Rao, S. Aruna Kumar, P. Jogi, K. Chinthan Bhat, B. Kuladeep Kumar, and P. Gouda, "Student placement prediction model: a data mining perspective for outcome-based education system," *International Jour-*

nal of Recent Technology and Engineering (IJRTE), vol. 8, no. 3, pp. 2497–2507, 2019.

[2] A. F. Meghji, N. A. Mahoto, Y. Asiri, H. Alshahrani, A. Sulaiman, and A. Shaikh, "Early detection of student degree-level academic performance using educational data mining," *PeerJ Computer Science*, vol. 9, p. e1294, 2023.

[3] M. Yagci, "Educational data mining: prediction of students' academic performance using machine learning algorithms," *Smart Learning Environments*, vol. 9, no. 1, p. 11, 2022.

[4] S. Batool, J. Rashid, M. W. Nisar, J. Kim, H.-Y. Kwon, and A. Hussain, "Educational data mining to predict students' academic performance: A survey study," *Education and Information Technologies*, vol. 28, no. 1, pp. 905–971, 2023.

[5] R. Asad, S. Arooj, and S. U. Rehman, "Study of educational data mining approaches for student performance analysis," *Technical Journal*, vol. 27, no. 01, pp. 68–81, 2022.

[6] D. Hooshyar, Y. Yang, M. Pedaste, and Y.-M. Huang, "Clustering algorithms in an educational context: An automatic comparative approach," *IEEE Access*, vol. 8, pp. 146 994–147 014, 2020.

[7] S. Križanić, "Educational data mining using cluster analysis and decision tree technique: A case study," *International Journal of Engineering Business Management*, vol. 12, pp. 1–9, 2020.

[8] M. Z. Hossain, M. N. Akhtar, R. B. Ahmad, and M. Rahman, "A dynamic k-means clustering for data mining," *Indonesian Journal of Electrical engineering and computer science*, vol. 13, no. 2, pp. 521–526, 2019.

[9] A. Qoiriah, R. Harimurti, A. I. Nurhidayat *et al.*, "Application of k-means algorithm for clustering student's computer programming performance in automatic programming assessment tool," in *International Joint Conference on Science and Engineering (IJCSE 2020)*. Atlantis Press, 2020, pp. 421–425.

[10] P. Tang, Y. Wang, and N. Shen, "Prediction of college students' physical fitness based on k-means clustering and svr," *Computer Systems Science and Engineering*, vol. 35, no. 4, pp. 237–246, 2020.

[11] M. Silva, R. Rupasingha, and B. Kumara, "A study for determining students' academic performance based on their activities using clustering approaches," 2022.

[12] S. J. Sultan Alalawi, I. N. Mohd Shaharanee, and J. Mohd Jamil, "Clustering student performance data using k-means algorithms," *Journal of Computational Innovation and Analytics (JCIA)*, vol. 2, no. 1, pp. 41–55, 2023.

[13] I. Gunawan, D. E. Kusumaningrum, T. Triwiyanto, W. Zulkarnain, A. Nurabadi, M. F. A. Sanutra, N. S. Rosallina, M. A. Rofiq, F. Afiantari, K. P. Supriyanto *et al.*, "Hidden curriculum and character building on self-motivation based on k-means clustering," in *2018 4th International Conference on Education and Technology (ICET)*. IEEE, 2018, pp. 32–35.

[14] Q. Wang, "Application of the intra cluster, characteristic of k-means clustering method in english score analysis in colleges," in *Journal of Physics: Conference Series*, vol. 1941, no. 1. IOP Publishing, 2021, p. 012001.

[15] A. Rauthan, A. S. Singh, N. Singh *et al.*, "Impact on higher education in pandemic: analysis k-means clustering using urban & rural areas," in *2021 3rd International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)*. IEEE, 2021, pp. 1974–1980.

[16] V. Bahel, S. Malewar, and A. Thomas, "student interest group prediction using clustering analysis: an edm approach," in *2021 International Conference on Computational Intelligence and Knowledge Economy (ICCIKE)*. IEEE, 2021, pp. 481–484.

[17] W. Purba, S. Tamba, and J. Saragih, "The effect of mining data k-means clustering toward students profile model drop out potential," in *Journal of Physics: Conference Series*, vol. 1007. IOP Publishing, 2018, p. 012049.

[18] O. J. Oyelade, O. O. Oladipupo, and I. C. Obagbuwa, "Application of k means clustering algorithm for prediction of students academic performance," *arXiv preprint arXiv:1002.2425*, 2010.

[19] A. M. Ikotun, A. E. Ezugwu, L. Abualigah, B. Abuhaija, and J. Heming, "K-means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data," *Information Sciences*, 2022.

[20] R. Nainggolan, R. Perangin-angin, E. Simarmata, and A. F. Tarigan, "Improved the performance of the k-means cluster using the sum of squared error (sse) optimized by using the elbow method," in *Journal of Physics:*

Conference Series, vol. 1361, no. 1. IOP Publishing, 2019, p. 012015.

[21] J.-O. Palacio-Niño and F. Berzal, "Evaluation metrics for unsupervised learning algorithms," *arXiv preprint arXiv:1905.05667*, 2019.

[22] Z. Gu, "Complex heatmap visualization," *Imeta*, vol. 1, no. 3, p. e43, 2022.